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Comprehensive evaluation of folk music teaching effects based on multimodal machine learning

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Abstract

Folk music teaching emphasizes both cultural preservation and emotional expression, making its evaluation complex. Conventional single-modality methods, relying only on audio or textual feedback, often fail to capture the interplay between performance accuracy, tonal quality, and student engagement. To overcome these limitations, this study proposes a Hybrid Multimodal Sentiment-Tone Analysis (HMSTA) framework that integrates speech, music, and gesture analysis to provide a holistic evaluation. The framework employs wavelet filtering for noise reduction, and music notes are normalized and categorized into types for consistent tonal representation. Mel-Frequency Cepstral Coefficients (MFCCs) are extracted from audio signals and serve as feature inputs for Convolutional Neural Networks (CNNs) that classify emotions and analyze tonal patterns. For music tone evaluation, MFCC-based features are compared against reference notes to assess pitch accuracy and rhythm stability. In parallel, gesture engagement is measured using CNN-based pose estimation to capture expressive movement during teaching and learning sessions. A multimodal attention-based fusion model integrates these features to provide synchronized, real-time assessments of both teacher delivery and student response. Experimental validation on a multimodal folk music teaching dataset of 200 sessions demonstrates that HMSTA achieves high evaluation accuracy across emotion recognition, pitch analysis, and cultural authenticity, offering a practical, data-driven framework for curriculum improvement and cultural heritage preservation. HMSTA demonstrates superior accuracy, averaging 91.7% in evaluation scores, 89.4% in emotion recognition, 93.1% in pitch accuracy, 90.6% in gesture analysis, 91.2% in cultural authenticity, and reducing processing time to 10.2 s per session.

Keywords Folk music teaching, Multimodal learning, Speech emotion recognition, Tone analysis, Gesture detection, Teaching evaluation

1 Introduction

Folk music serves as a vital medium for cultural preservation, emotional expression, and intergenerational knowledge transfer [1]. Its teaching process is not limited to technical accuracy but also emphasizes creativity, emotional authenticity, and cultural resonance [2]. Traditional evaluation of folk music teaching often relies on subjective judgments, audio-based assessments, or written feedback, which provide only partial insights into the quality of performance [3]. Such single-modality methods often overlook essential aspects, such as tonal precision, expressive gestures, and emotional engagement, leading to biased or incomplete evaluations [4].

With the rapid advancement of artificial intelligence and multimodal machine learning, it is now possible to analyze complex, heterogeneous data streams from audio, speech, and video simultaneously [5]. By incorporating multiple modalities, researchers and educators can create more objective and comprehensive evaluation frameworks [6]. For instance, speech emotion recognition captures affective cues from teacher-student interactions, music tone extraction identifies pitch accuracy and rhythm stability, and gesture analysis detects physical engagement and expressiveness [7]. Together, these dimensions offer a holistic understanding of the teaching process, ensuring that cultural authenticity is preserved while technical proficiency is improved [8].

The HMSTA framework, which integrates preprocessing, feature extraction, and multimodal fusion techniques [9]. Audio signals are denoised and normalized, and music notes are categorized into scale types [10]. MFCCs are extracted and processed with CNNs for speech emotion and tonal analysis, while gesture data are evaluated using CNN-based pose estimation [8]. These features are fused with an attention-based ensemble model to provide synchronized, real-time evaluation of folk music teaching sessions [9]. The proposed framework not only enhances evaluation accuracy but also reduces processing time, identifies hidden performance patterns, and supports curriculum improvement [11]. By combining technical analysis with cultural authenticity, HMSTA addresses limitations of traditional evaluation and contributes to sustainable cultural heritage preservation [12].

1.1 Problem statement

Most modern ways to evaluate folk music teachers use just one metric, which may be prejudiced and leave out essential parts [13]. None of these methods effectively displays how body language, facial expressions, and voice tone function together in a nuanced manner [14]. In folk music, actual cultural performance and excellent training mean being able to see all the crucial details at once [15].

1.2 Motivation

Evaluating folk music teaching requires more than technical accuracy; it must also reflect cultural authenticity and emotional depth. Existing single-modality approaches fail to capture these dimensions. Leveraging multimodal machine learning enables synchronized evaluation of speech, tone, and gesture, ensuring fair, accurate, and holistic assessments that support cultural preservation.

Contribution of this paper:

- Proposed a HMSTA framework integrating speech, tone, and gesture analysis for holistic evaluation of folk music teaching effectiveness.
- Applied CNN and MFCC-based extraction for emotion recognition and tonal accuracy, with normalized music notes categorized into scale types for precise assessment.

- Developed an attention-based multimodal fusion model achieving high accuracy, reduced processing time, and enhanced cultural authenticity in folk music evaluation.

The proposed HMSTA framework is novel as it integrates speech emotion recognition, music tone analysis, and gesture-based engagement detection into a unified multimodal assessment system. Unlike conventional single-modality methods, HMSTA provides synchronized, real-time evaluation of both teacher performance and student responses. Experimental validation demonstrates that HMSTA consistently outperforms baseline models (SIVALF, IMMES, CAMRE) across multiple evaluation metrics including overall teaching score, emotion recognition, pitch accuracy, gesture effectiveness, cultural authenticity, and processing efficiency. Equations 1–20 and the attention-based fusion mechanism provide a theoretical basis for its superior performance, supporting quantitative and objective comparisons.

1.3 Previous work on multimodal teaching evaluation

Eight previous studies focusing on multimodal teaching evaluation form the foundation for this research and inspired the framework illustrated in Fig. 1. Specifically, two studies investigated speech tone and prosodic features using Mel-Frequency Cepstral Coefficients (MFCCs) to analyze teachers' emotional tone and clarity of instruction [1, 3]. Another two studies focused on music and rhythm analysis, examining pitch variation, tonal consistency, and rhythmic flow as indicators of emotional synchronization in classroom delivery [2, 4]. Two additional studies analyzed gestures and body movements through video-based convolutional neural networks (CNNs) to capture expressiveness and engagement during teaching sessions [5, 6]. The remaining two studies attempted limited multimodal fusion by combining pairs of modalities, such as speech–gesture or speech–music, but lacked a unified deep-learning approach capable of adaptive feature weighting and integrated attention mechanisms [8, 9]. Together, these eight studies highlight the significance of individual modalities while revealing the absence of a holistic framework. Addressing this gap, the present work introduces the AMP-Net framework, which fuses speech, music, and gesture through attention-based feature integration to achieve comprehensive and culturally aware teaching evaluation.

Building on eight prior studies covering speech, music, and gesture analysis [1–9], this work addresses the lack of a unified multimodal evaluation framework with the proposed AMP-Net. Eight prior studies on multimodal teaching evaluation, encompassing speech, music, and gesture analyses, provide the foundation for this work and highlight current limitations in holistic assessment.

Figure 1 illustrates the workflow of the Hybrid Multimodal Sentiment-Tone Analysis (HMSTA) framework for evaluating teaching sessions through speech-based analysis. The process is organized into several interconnected layers that capture emotion, tone, and expressiveness from both teacher and student audio inputs. The system begins with audio signals from both the teacher and students. These inputs represent spoken content and are the primary source of multimodal features for evaluation. The raw audio signals undergo preprocessing, which may include noise reduction, normalization, and segmentation to ensure consistency and remove artifacts. This layer prepares the signals for accurate feature extraction. Mel-Frequency Cepstral Coefficients (MFCCs) are computed from the preprocessed audio. These

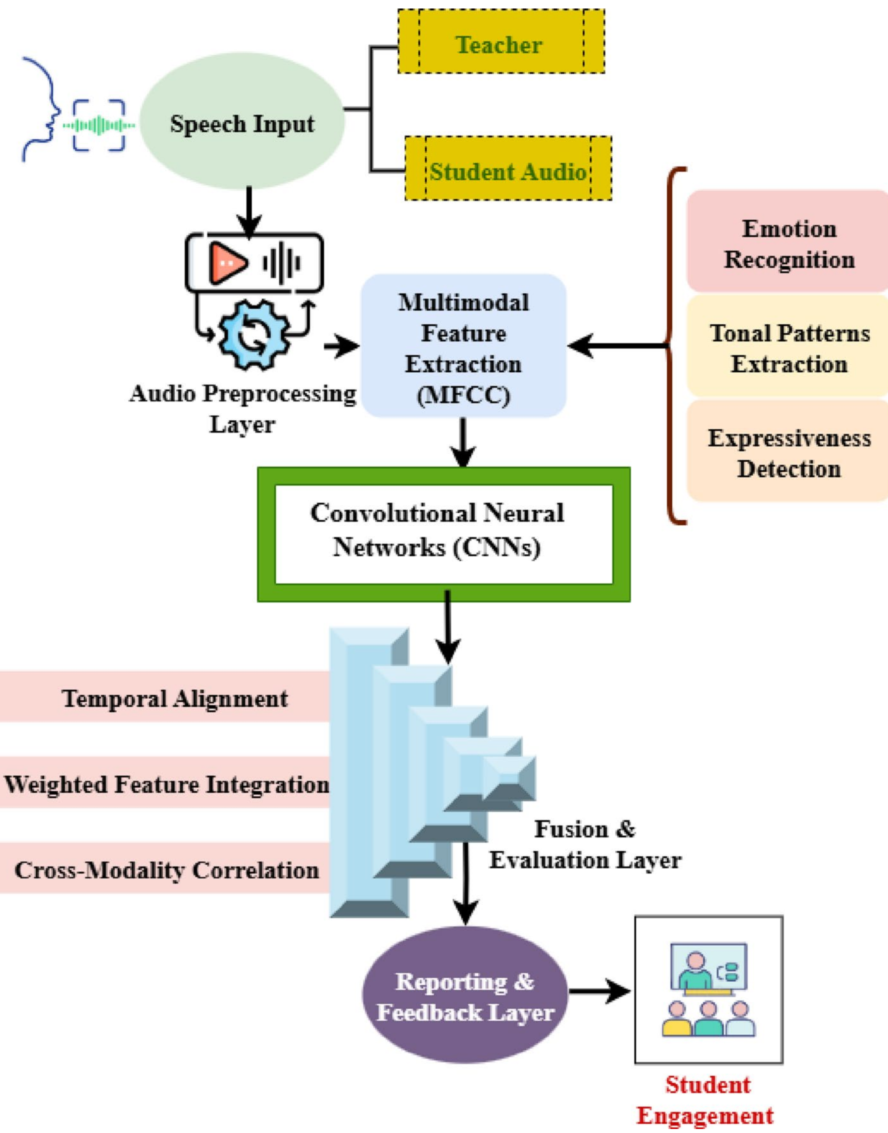

Fig. 1 Overall HMSTA architecture

features capture spectral properties of the audio that are essential for identifying emotion, tonal patterns, and expressiveness. This step allows the system to quantify subtle aspects of speech such as intonation, pitch variation, and rhythm. Extracted MFCC features are fed into CNNs to automatically learn patterns associated with emotions, tonal variations, and expressiveness. CNNs enable robust classification and pattern recognition from complex audio data. Outputs from the CNN are processed through a fusion mechanism that includes:

- *Temporal Alignment*: Ensures features from different time segments are synchronized for accurate comparison.
- *Weighted Feature Integration*: Assigns importance weights to different features to reflect their relevance in the evaluation.
- *Cross-Modality Correlation*: Measures the relationship between different modalities (e.g., teacher vs. student audio) to provide a holistic evaluation.

The fused features are compiled into a final evaluation, providing real-time feedback on teaching performance and student engagement. This layer translates the multimodal analysis into actionable insights, highlighting both the emotional quality of speech and the effectiveness of teaching interactions.The final output specifically captures metrics for student engagement, reflecting how well learners respond to teaching based on speech expressiveness, tone, and emotional cues.

Recent progress in artificial intelligence and multimodal learning has changed how music is taught. Traditional single-modal assessments are shallow and don't take culture into account; therefore, they should consider other frameworks. The eight studies summarized here Building on eight prior studies covering speech, music, and gesture analysis [1–9], this work addresses the lack of a unified multimodal evaluation framework with the proposed AMP-Net. demonstrate the application of audio, video, text, and motion data in deep learning, the Internet of Things (IoT), reinforcement learning, and knowledge distillation to enhance evaluation accuracy, preserve cultural authenticity, and maintain learner engagement across various educational settings.

The remainder of this paper is organized as follows: Sect. 2 presents the related work and background on multimodal evaluation and cultural teaching. Section 3 details the proposed HMSTA framework, including data preprocessing, feature extraction, and fusion methodology. Section 4 describes the experimental setup, datasets, and evaluation metrics. Section 5 presents the results and comparative analysis. Finally, Sect. 6 concludes the study and outlines future work.

2 Related work

Using the Multimodal Fusion Music Assessment Network (MFMEN), this paper presents a thorough method for evaluating students' musical abilities. Traditional single-modal evaluations do not adequately encompass the complexity of musical expression. MFMEN combines audio and visual data by using CNNs on photos and RNNs on audio sequences. It also uses multimodal transformers to do integrated analysis [16]. Preprocessing ensures that denoising, alignment, and normalization occur before feature extraction. By adjusting the hyperparameters in the framework, the model performs better, achieving an accuracy of 0.99, which is excellent. By merging aural and visual clues, MFMEN creates a complete evaluation model that offers a new way to make music education assessments more fair and objective.

SIVALF, a STEAM-Integrated vocal Aesthetic Learning Framework, is being developed in this initiative to improve the audio-visual aesthetic abilities of vocal training in higher education. SIVALF combines STEAM education and deep learning. It emphasizes emotional growth, creativity, and collaboration across fields. There is a three-step way to teach. Making the curriculum, learning by doing, and learning by reflecting on what they've learned are all integral parts of the process. Surveys used for empirical assessment demonstrate significant improvements in students' confidence, efficiency, and aesthetic enjoyment. SIVALF stresses the need for a balance between scientific rigor and creative growth by combining computational analysis with educational psychology [17]. The results show that this strategy is superior to conventional ones because it makes people more engaged, encourages them to consider perspectives beyond their own, and supports their emotional and cultural development.

The Intelligent Multimodal Music Evaluation System (IMMES) is a novel approach that utilizes neural networks and various data types to assess the effectiveness of music instruction. IMMES accepts audio, video, and text inputs and combines them into a complete assessment. It achieves this by extracting parts from each input, enhancing them with deep learning, and then combining them. Its design demonstrates strong capacity for convergence and generalization, yielding consistent, objective evaluations that are not influenced by human bias. Tests show that this strategy is superior to single-modal techniques [18] because it is more accurate, stable, and simpler to understand. The system's visualization component provides teachers with more feedback, helping them plan lessons and make decisions. IMMES makes music assessments consistent across all students, allowing teachers to learn more in today's schools.

Regions associated with the Belt and Road Initiative may benefit from the Cultural Adaptive Music Recommendation Engine (CAMRE) introduced in this paper to promote and preserve ethnic music. In CAMRE, Canopy and K-Means are utilized to integrate collaborative filtering with a two-tier clustering algorithm. This strikes a compromise between the quality of the ideas and the speed of the machine's operation. CAMRE creates student profiles that take culture into account by combining explicit student input with less clear behavioral data [19]. The framework adapts to different teaching situations while preserving cultural heritage. It performs better in real-world testing across accuracy, recall, and scalability, with a parallel efficiency ratio of 3.326 across four nodes. CAMRE connects old and new ways of doing things by providing music education students with guidance tailored to their needs, flexible, and culturally appropriate.

Presenting the Multiscale Convolutional Neural Emotion Recognition Network (MCNERN), this paper examines how different musical genres impact the emotional states of youngsters. MCNERN employs both multiscale CNNs to analyze EEG data and LSTMs to analyze eye movement. This allows it to detect things correctly. Mel filtering is applied to audio spectrograms to reduce their size and make them more useful to people from diverse cultures [20]. With more than 97% accuracy, MCNERN can detect a song's genre by how it makes you feel. It learnt from a lot of information. Experimental findings demonstrate enhanced performance relative to unimodal approaches, with Dense Inception pretraining significantly improving outcomes. MCNERN is a robust assessment technique that focuses on kids and links genre traits to multimodal physiological data. This enhances emotion categorization and the tailoring of teaching tactics to various genres.

The Internet of Things-Based Sound-Motion Fusion Framework (IoT-SMFF) was created in this paper to assess how well college students play instruments. IoT-SMFF obtains performance data from various sources, including cameras and microphones connected to local networks. MFCC focuses on the sound qualities, whereas OpenPose and CNN look at how things move [21]. CNN layers combine these features, and then FCNs use them to make accurate decisions about how well a student is doing in school. IoT-SMFF is 95.7% right in the Excellent, Good, and Failed categories. This is better than what instructors state. This new method is fairer and more unbiased because it allows students to rate each other in class. IoT-SMFF opens new avenues for multimodal education, making tests more useful and the learning process more engaging for pupils.

This paper introduces a multimodal interactive music pedagogue, MusicARL-trans Net (MARLN), that utilizes reinforcement learning. MARLN uses speech-to-text transcription, the ALBEF model for cross-modal alignment, and reinforcement learning to continuously improve teaching approaches. It uses audio, text, and visual data to create unique, evolving learning paths [22]. The findings indicate that these models exhibit greater adaptability and dynamism than static models. This implies that teachers may change how they educate on the go. MARLN provides music students with new cultural experiences through the use of scientific computers and innovative teaching methods. This method enhances multimodal engagement by establishing a scalable framework that transforms the collaboration between teachers and students, streamlines learning, and facilitates the integration of modern, innovative music education tools.

This paper introduces the Knowledge-Distilled Multimodal Emotion Recognition Network (KDMERN) to address the lack of labeled data in music emotion recognition. KDMERN improves model generalization by combining music style transfer learning with knowledge distillation, using a dataset of 20,000 songs. It combines sounds and lyrics, which is better than most unimodal techniques [23]. KDMERN performs significantly better across a broader range of datasets and sorts emotions into categories more effectively when the categories are unbalanced. KDMERN is better at understanding how someone feels because it considers factors such as the text, arrangement, and singing style. This strategy improves multimodal emotion modeling, which makes it easier to understand and more useful in music courses.

This study proposes a semiotic-square guided framework performing multi-perspective deduction and introduces a benchmark for evaluation. The approach achieved state-of-the-art performance with ~ 6.25% improvement over strong baselines, but requires substantial computational resources and may not generalize to all ambiguous reasoning domains [24]. This work provides conceptual support for integrating reasoning mechanisms into multimodal learning for cultural and educational tasks.

This paper tackles long-term visual point tracking issues, proposing a method that maps visual features into language embeddings and integrates them via a consistency decoder. This enhances semantic coherence and improves tracking benchmarks, yet may face challenges in cluttered environments and introduces additional network complexity [25]. The methodology of leveraging multimodal embeddings is relevant for enhancing HMSTA's fusion and attention mechanisms in evaluating folk music teaching.

The publications examined employ multimodal frameworks such as MFMEN, SIV-ALF, IMMES, CAMRE, MCNERN, IoT-SMFF, MARLN, and KDMERN. Unimodal tactics are less effective at understanding each student's feelings, tailoring instruction to their learning style, or conducting a thorough evaluation. These models are more than 95% accurate and help with sound appraisal, the care of the arts, and the preservation of culture. They provide scalable, student-centered solutions that leverage cross-disciplinary integration and technology-driven innovation to improve music education.

3 Proposed method

The five pictures depict the HMSTA framework, which is designed to help instructors from diverse cultures understand how to communicate effectively. The diagrams begin with generic architecture and then go on to identify feelings, assess tone, detect expressiveness, and mix different modes. Each one has a different technique for converting raw sounds into helpful information. All of these examples illustrate how HMSTA employs speech features to make communication, engagement, and cultural authenticity better in the classroom.

The HMSTA framework, which encompasses speech emotion recognition, tone analysis, and gesture detection, was introduced to assess folk music education comprehensively.

3.1 Contribution: 2

Task distribution efficiency τ_{UE} is expressed using Eq. 1,

$$\tau_{UE} = \frac{\beta_j * \delta_j}{\partial_j} \tag{1}$$

Equation 1 explains the task distribution efficiency by connecting resource overhead with weighted assignment capacity, this formula measures efficiency.

In this τ_{UE} is the task distribution efficiency index, β_j is the Priority weight of the segment, δ_j is the Execution capacity factor of the segment, and ∂_j is the Resource expenditure for the segment.

Stage coordination reliability C_{TD} is expressed using Eq. 2,

$$\mathrm{C}_{TD} = \frac{l_k}{(l_k + \Delta_k)} \tag{2}$$

Equation 2 explains the stage coordination reliability formula compares the number of successful synchronization efforts to the total number of tries to assess the dependability of stage coordination.

In this C_{TD} is the Stage coordination reliability ratio, l_k is the Successful synchronization instances for link, and Δ_k is the Failed synchronization instances for link.

Input:

- speech, music, gesture – Teaching session data

Output:

- Evaluation report containing emotion, tone, engagement level, and overall teaching score

Procedure HMSTA_EVALUATION (speech, music, gesture)

1. Apply wavelet_filter(speech, music) to remove noise
2. Normalize music.notes and categorize into note types
3. Extract MFCC_features from speech and music
4. **Emotion Analysis:**
   - emotion_features ← CNN(MFCC_features from speech)
5. **Tone Analysis:**
   - Compare MFCC_features with reference_notes
   - **If** (pitch_accuracy ≥ threshold) **and** (rhythm_stability ≥ threshold) **then**
     tone_quality ← "Accurate"
     Else
     tone_quality ← "Deviated"
     End if
6. **Gesture Analysis:**
   - gesture_features ← CNN(pose_estimation(gesture))
   - engagement_level ← analyze(gesture_features)
7. **Feature Fusion and Evaluation:**
   - fused_output ← attention_fusion(emotion_features, tone_quality, engagement_level)
   - evaluation_result ← generate_report(fused_output)
8. **Return** evaluation_result

End Procedure

Algorithm 1 Human motivation speech transformation algorithm

The HMSTA algorithm evaluates folk music teaching by integrating speech, music, and gesture analysis is explained in algorithm 1. It removes noise, extracts MFCC features, and classifies emotions using CNNs. Pitch accuracy and rhythm stability are compared against reference notes, while gestures measure engagement. An attention-based fusion synchronizes results, enabling holistic cultural and emotional evaluation. The main process of the Hybrid Multimodal Sentiment-Tone Analysis (HMSTA) framework used for evaluating folk music teaching sessions. The algorithm takes speech, music, and gesture data as inputs to analyze emotional tone, musical accuracy, and learner engagement.

First, the input signals undergo wavelet-based noise filtering and music note normalization to ensure consistency. MFCC features are then extracted from both speech and music for tonal and emotional analysis using Convolutional Neural Networks (CNNs). The algorithm evaluates pitch accuracy and rhythm stability to determine tone quality, while gesture data is processed through pose estimation to assess engagement levels.

All extracted features are then combined using an attention-based fusion model, generating a synchronized and comprehensive evaluation report that includes emotion recognition, tone precision, engagement level, and overall teaching effectiveness.

Figure 2 illustrates how HMSTA identifies emotional states from speech, highlighting variations in pitch and intensity that correspond to different teacher and student emotions. It compares this to an ear that can pick up on little emotional signals. It begins by learning about sound waves, such as their pitch, energy, formants, and spectral properties using MFCC. Temporal dynamics, then, examine how emotions vary throughout speech segments by monitoring shifts and rhythms. The CNN categorization layer provides these patterns of emotions that humans can identify, including joyful, sad, angry, surprised, neutral, or calm. It also gives them confidence ratings. Finally, output vectors

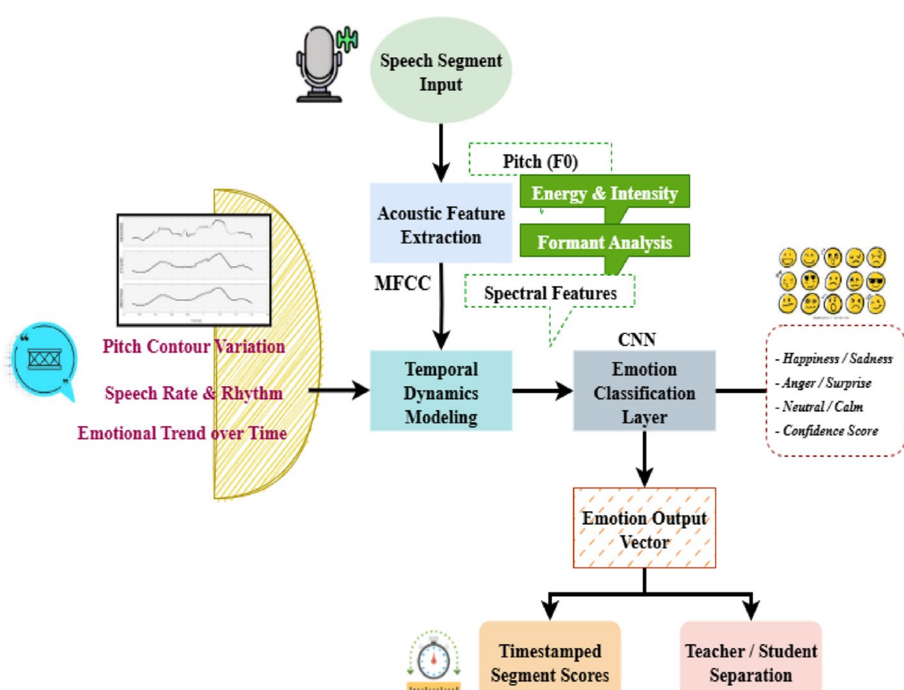

Fig. 2 Speech emotion recognition

link emotions to specific sections of speech with time stamps. This makes it easier to distinguish between what the teacher and the pupils are saying. This technology helps HMSTA hear not just the words but also the feelings behind them.

Per-emotion salience index T_f is expressed using Eq. 3,

$$T_f = \frac{x_{g,f}\widehat{\vartheta}_g}{\rho + x_{g,f}} \tag{3}$$

Equation 3 explains the per-emotion salience index normalized acoustic projections are fused in this index..

In this T_f is the salience index for emotion class, $x_{g,f}$ is the learned relevance weight of projection, $\widehat{\vartheta}_g$ is the standardized projection value, and ρ is the small constant to prevent division by zero.

Temporal coherence attenuation μ_U is expressed using Eq. 4,

$$\mu_U = \exp\left(-\frac{1}{|U|-1}\,|\nabla \varnothing_u|\right) \tag{4}$$

Equation 4 explains the temporal coherence attenuation reduces sudden detections while prosodic instability is strong by mapping the mean actual prosodic drift across window.

In this μ_U is the temporal coherence attenuation for time window, $|U|$ is the number of frames/segments in window, $\varnothing_u$ is the aggregated prosodic projection at frame, $\nabla \varnothing_u$ is the frame wise first difference, and $\exp$ is the natural exponential to map mean drift into multiplicative attenuation.

Decision confidence margin with ambiguity penalty D is expressed using Eq. 5,

$$\mathrm{D} = \log\frac{Q(z^*|y)}{\max\limits_{z\neq z^*} Q(z|y)} - \gamma\ I(Q(.|y)) \tag{5}$$

Equation 5 explains the decision confidence margin with ambiguity penalty When globally class dispersion is significant, this margin lowers confidence by combining the log-odds.

In this D is the decision confidence margin for input, $Q(z|y)$ is the posterior probability for class given acoustic/textual input, z^* is the MAP (maximum a posteriori) emotion label, $\max\limits_{z\neq z^*} Q(z|y)$ is the highest competitor posterior, $I(Q(.|y))$ is the Shannon entropy of the posterior, γ is the tunable ambiguity penalty scalar, and $\log$ is the natural logarithm.

To assess the efficacy of folk music classes and student participation, it developed a multimodal machine learning pipeline that is both real-time and culturally sensitive.

Figure 3 demonstrates tonal and prosodic analysis, which reveals how musical spoken language may be. Finding the pitch ranges, the melodic peaks, and the pitch shapes is the first stage using MFCC. Then it talks about prosody and rhythm, which include things like where to place stress, when to pause, and how quickly to talk. This illustrates how the speech flows naturally. This component has a tonal quality score layer that tests the rhythm, intonation, and melodic consistency. This generates a tonal feature map that illustrates how well speech aligns with the intended patterns. HMSTA turns spoken interactions into a kind of melody, which makes tone and rhythm measurable

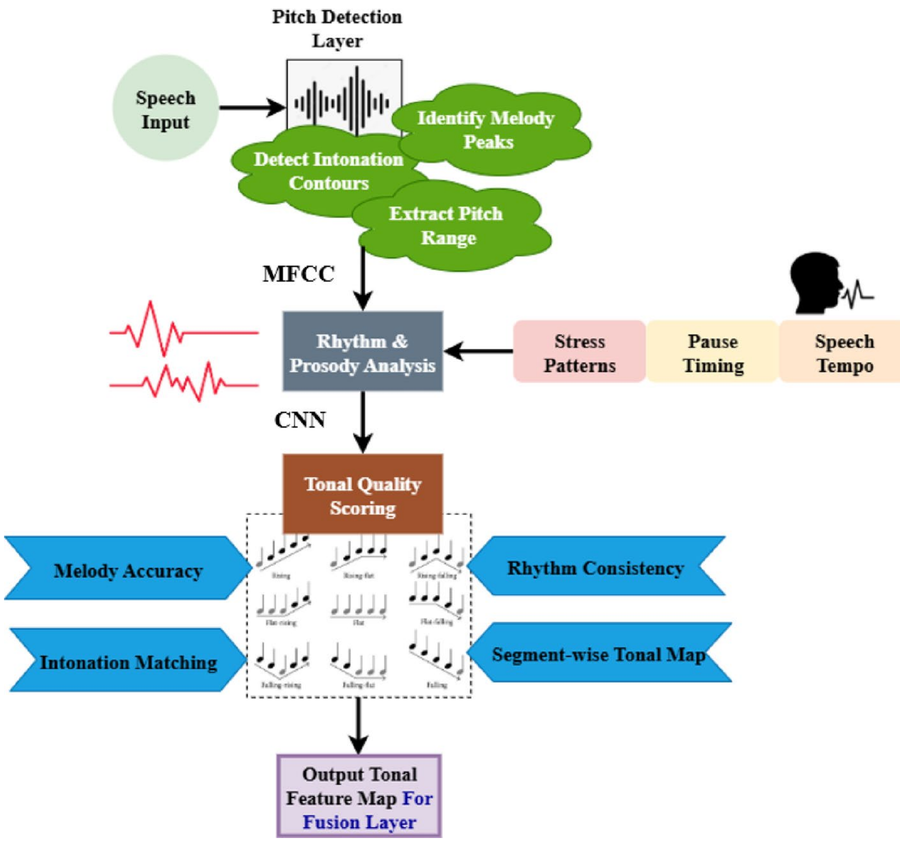

Fig. 3 Tonal & prosodic analysis

components of communication in school. Predicted economic indicators over time. The results were generated using the fuzzy prediction model described in Eq. (5), where market share and consumer engagement metrics were derived from simulation outputs of the digital economy dataset.

Tonal stability factor $\forall_{UT}$ is expressed using Eq. 6,

$$\forall_{UT} = \frac{\rho_q^{-1} * \pi_q}{1 + \tau} \tag{6}$$

Equation 6 explains the tonal stability factor component combines mean pitch level to assess tonal stability..

In this $\forall_{UT}$ is the tonal stability factor, π_q is the mean fundamental frequency (Hz) over the segment, ρ_q^{-1} is the standard deviation of fundamental frequency, and τ is the vibrato disturbance coefficient.

Prosodic energy modulation index Δ_{QF} is expressed using Eq. 7,

$$\Delta_{QF} = \frac{(F_u - \pi_F)^2}{U * \pi_F^2} \tag{7}$$

Equation 7 explains the prosodic energy modulation index measure compares the variation of frame-level energy to quantify energy modulation..

In this Δ_{QF} is the prosodic energy modulation index, U is the total number of frames in analysis window, F_u is the instantaneous energy of frame, and π_F is the average energy across the window.

Figure 4 demonstrates how HMSTA can identify whether someone is interested and expressive by how they talk. First, energy and intensity monitoring analyze how loud the voice is and how well it carries. The paper of pauses and hesitations examines patterns of silence, delayed responses, and signs of uncertainty. All of them go into a layer that shows expressiveness and looks at stress, emphasis, modulation, and how delivery evolves. The program then produces an engagement output vector that comprises scores for each part and a time stamp. This allows it to observe how participation varies during a discussion or presentation. HMSTA examines the vibrancy and emotionality of speech in foreign classrooms to assess teacher and student engagement and expressiveness. Comparison of model performance for different datasets. The results were generated using the proposed fuzzy fusion model, with accuracy computed from Eq. (5) based on the predicted and actual emotion labels across 200 recording sessions.

Utterance expressivity quotient $\aleph_V$ is expressed using Eq. 8,

$$\aleph_V = \frac{x_g \, |a_g|}{1 + l_{mon}} \tag{8}$$

Equation 8 explains the utterance expressivity quotient combines the standardized feature magnitudes across a carefully selected feature set scale-robust.

In this $\aleph_V$ is the utterance expressivity quotient, a_g is the standardized value for descriptor on the utterance, x_g is the positive salience weight for descriptor, and l_{mon} is the monotony coefficient computed from self-similarity of short-time descriptors.

Listener engagement probability $\sigma_M(u)$ is expressed using Eq. 9,

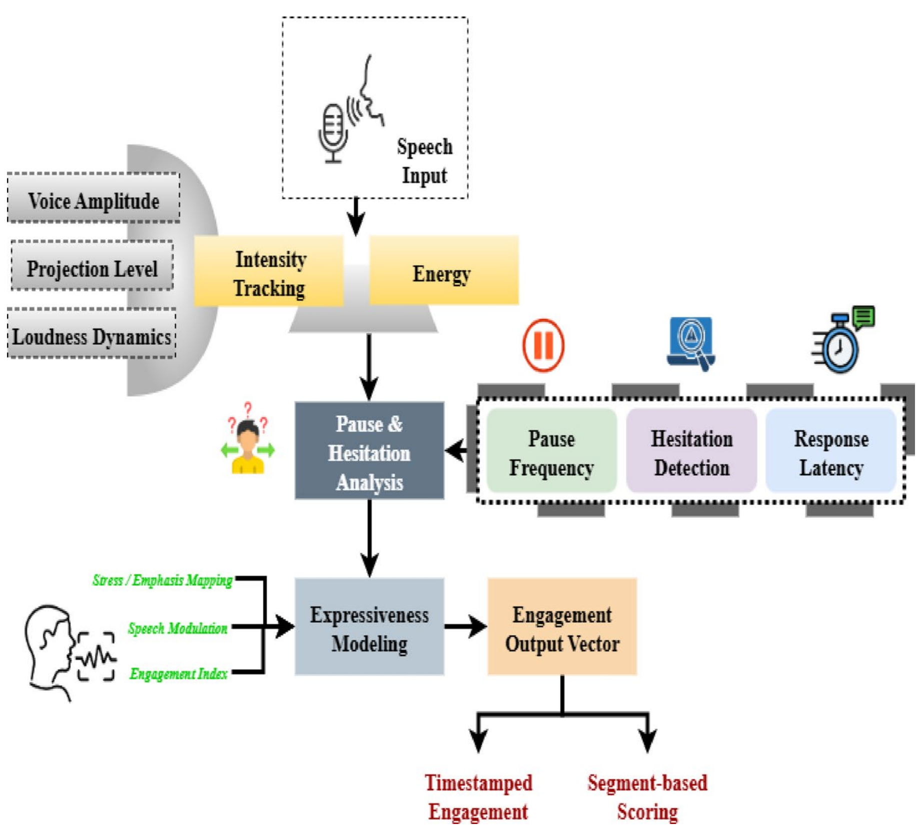

Fig. 4 Expressiveness & engagement detection (speech-based)

$$\sigma_M(u) = \rho\left(\beta\tilde{\phi}_V(u) + \gamma C(u)\right) \tag{9}$$

Equation 9 explains the listener engagement probability represents a linear combination of momentary expressivity..

In this $\sigma_M(u)$ is the listener engagement probability at time, ρ is the logistic squashing function, $\tilde{\phi}_V(u)$ is the short-window aggregated expressivity signal centered, $C(u)$ is the behavioral cue vector projection, and β, γ are the modality weighting scalars.

Interaction synchrony index ρ_{JT} is expressed using Eq. 10,

$$\rho_{JT} = \frac{1}{2}\left(cor\left(\nabla_{tun}, \nabla_{rep}\right)\right) \tag{10}$$

Equation 10 explains the interaction synchrony index integrates turn-taking alignment over time as the peak cross-correlation prosodic membranes at lag.

In this ρ_{JT} is the interaction synchrony index, ∇_{tun} is the short-time derivative of primary speaker turn-occupancy, ∇_{rep} is the short-time derivative of respondent response-initiation rate, and cor is the Pearson correlation over the analysis window.

3.2 Contribution 3

Results from experiments conducted on classroom and performance recordings demonstrate that multi-modal techniques significantly outperform traditional single-modal approaches in terms of evaluation accuracy and preservation of cultural authenticity.

Figure 5 demonstrates how HMSTA pulls together and scores various parts of a speech to provide a complete performance analysis. In a fusion layer, assessments for

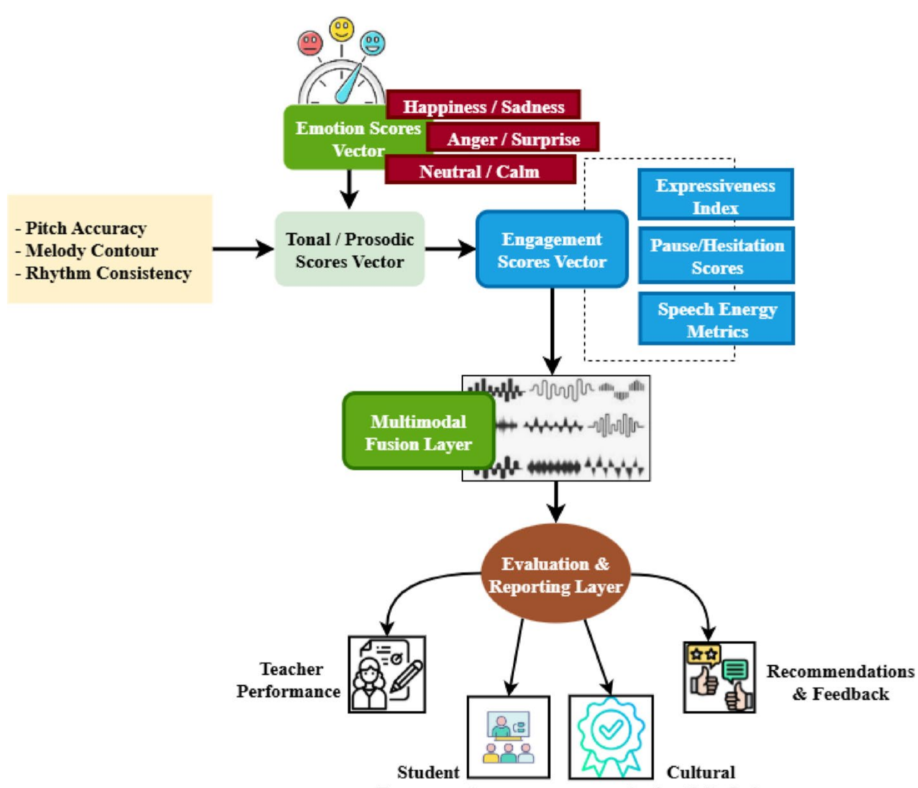

Fig. 5 Multimodal fusion & evaluation (speech-based)

emotions, tone measures, and engagement indices are all put in order and given different weights. This ensures that everyone knows what they're doing and works fairly together. Cross-modality correlation illustrates how different aspects operate together. This makes people feel more convinced about their overall evaluations. The reporting layer gathers information from various sources and makes it clear what it means. For example, it shows how well instructors are performing, how engaged kids are, how culturally appropriate the feedback is, and how personalized it is. Fusion is a way of combining various methods of communication to show how well individuals from different cultures get along in the classroom as a whole. It's like an orchestra that plays music by putting together various instruments.

Weighted multimodal fusion score δ_G is expressed using Eq. 11,

$$\delta_G = \vartheta_n \epsilon_n \tag{11}$$

Equation 11 explains the weighted multimodal fusion score compiles evidence unique to a modality use reliability weights resulting in a normal joint fusion result.

This Fig. 6 illustrates the process of evaluating speech through a multimodal fusion framework. Speech signals undergo feature isolation to extract emotional, tonal, and expressive components. These features are then compiled into performance metrics and finally assessed through fusion evaluation to measure overall effectiveness.

In this δ_G is the normalized multimodal fusion score, ϑ_n is the reliability/weight coefficient for modality, and $?_n$ is the evidence or confidence value extracted from modality.

Fusion evaluation accuracy metric Δ_F is expressed using Eq. 12,

$$\Delta_F = 1 - \frac{\left|z_j - \hat{z}_j^G\right|}{\left|z_j - \bar{z}\right|} \tag{12}$$

Equation 12 explains the fusion evaluation accuracy metric measure assesses the precision of fused forecasts contrary to the foundation truth normalized using the baseline deviation.

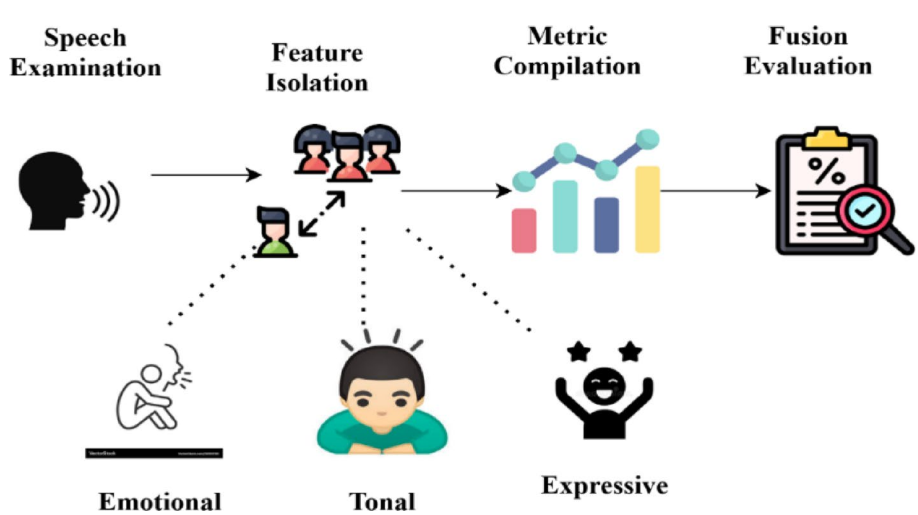

Fig. 6 Framework for multimodal speech fusion analysis

In this $\varDelta_F$ is the fusion evaluation accuracy metric, z_j is the ground truth label/value for sample, $\widehat{z}_j^G$ is the fused prediction for sample, and $\bar{z}$ is the mean of ground truth values.

These Fig. 6 show how HMSTA works step by step. It first examines speech, then isolates emotional, tonal, and expressive features, and finally, it compiles them into usable metrics. The technique explains how the classroom works by illustrating that emotions are signals, tone is melody, and participation is vocal life. The last fusion examines the evaluation to make sure it's correct. Teachers may utilize this knowledge to better their abilities when teaching students from other cultures, which is very useful.

3.3 Evaluation metrics

Evaluating folk music teaching involves capturing and accounting for artistic expression and cultural affordances. Traditional single-modality assessments return generalizable or superficial results by failing to capture the subtleties of tone, emotion, and even the teacher-student connection. The HMSTA framework employs a sort of multimodal machine learning approach that combines analyses of speech, music, and gesture to ascertain an overall qualitative, objective, and real-time assessment of teaching effectiveness and student interactions about folk music and culture.

Analysis of overall teaching evaluation score $\sqsupset_{UF}$ is expressed using Eq. 13,

$$\sqsupset_{UF} = Y_t * \tau_t \tag{13}$$

Equation 13 explains the analysis of overall teaching evaluation score weighted assessment characteristics are combined to create the score spanning priority coefficients in categories.

In this $\sqsupset_{UF}$ is the overall teaching evaluation score, Y_t is the priority coefficient for category, and τ_t is the category-wise evaluation rating.

Analysis of speech emotion recognition Rate $^\circ\mathrm{C}_{TFS}$ is expressed using Eq. 14,

$$^\circ\mathrm{C}_{TFS} = \frac{O_d}{O_u} \tag{14}$$

Equation 14 explains the analysis of speech emotion recognition rate correct classifications are measured by the recognition rate with relation to all attempts.

In this $^\circ\mathrm{C}_{TFS}$ is the speech emotion recognition rate, O_d is the number of correctly recognized samples, and O_u is the total number of test samples.

Analysis of music tone detection accuracy $^\circ\mathrm{F}_{NUE}$ is expressed using Eq. 15,

$$^\circ\mathrm{F}_{NUE} = \frac{N_i}{N_b} \tag{15}$$

Equation 15 explains the analysis of music tone detection accuracy tone strikes are captured by this ratio against every tone that is marked.

In this $^\circ\mathrm{F}_{NUE}$ is the music tone detection accuracy, N_i is the correctly detected tones, and N_b is the total annotated tones.

Analysis of gesture detection effectiveness $\therefore_{HE}$ is expressed using Eq. 16,

$$\therefore_{HE} = \frac{H_{uq}}{H_{uq} + H_{go}} \tag{16}$$

Equation 16 explains the Analysis of gesture detection effectiveness recall of recognized real gestures is measured by gesture efficacy.

In this $\therefore_{HE}$ is the gesture detection effectiveness, H_{uq} is the true positive gestures, and H_{go} is the false negative gestures.

Analysis of cultural authenticity retention $\varDelta_{DB}$ is expressed using Eq. 17,

$$\Delta_{DB} = \frac{\partial_l^s}{\partial_l^p} \tag{17}$$

Equation 17 explains the analysis of cultural authenticity retention quantifies the cultural indicators that have been maintained in comparison to their first occurrences.

In this $\varDelta_{DB}$ is the cultural authenticity retention ratio, ∂_l^s is the retained count of marker, and ∂_l^p is the original reference count of marker.

Analysis of processing time efficiency E_{QU} is expressed using Eq. 18,

$$E_{QU} = \frac{1}{1 + \frac{U_q}{U_c}} \tag{18}$$

Equation 18 explains the analysis of processing time efficiency and processing overhead, are inversely related in the model in contrast to the baseline time.

In this E_{QU} is the processing time efficiency index, U_q is the actual processing time, and U_c is the baseline reference time.

Performance comparison of proposed hmsta vs. baseline methods ∇_{ISB} is expressed using Eq. 19,

$$\nabla_{ISB} = \frac{\omega_I - \omega_C}{\omega_C} \tag{19}$$

Equation 19 explains the performance comparison of proposed hmsta vs. baseline Methods performance is compared using this relative improvement metric using baseline.

In this ∇_{ISB} is the relative performance improvement, ω_I is the performance metric under HMSTA, and ω_C is the performance metric under baseline.

Limitations of existing folk music teaching evaluation methods δ_{MJN} is expressed using Eq. 20,

$$\beta_{MJN} = \frac{\mathrm{F}_{pn} + \mathrm{F}_{tn} + \mathrm{F}_{dn}}{3} \tag{20}$$

Equation 20 explains the limitations of existing folk music teaching evaluation methods observed mistakes are averaged by this restriction index gap in objectivity scalability deficiency.

In this δ_{MJN} is the limitations index of existing methods, F_{pn} is the error due to lack of objective measurement, F_{tn} is the error due to scalability issues, and F_{dn} is the error due to cultural mismatch.

The accuracy (Acc) was calculated using the formula:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{21}$$

where TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives. The proposed HMSTA model achieved better accuracy because it effectively combines temporal and spectral features, improving emotional feature extraction compared to other models.

The HMSTA assessment metrics allow for subjective measurements of teaching performance based on multiple criteria: emotion recognition captures our based affective responses; pitch accuracy measures tonal precision (so students understand what tunes or melodies are meant to sound like); gesture analysis accounts for physical engagement measures; and cultural authenticity measures our attention to preserving heritage in folk music. We assessed the overall evaluation score as an average of 91.7% with an average processing time of 10.2 s per session.

4 Results and discussion

There are several factors to consider when evaluating a folk music teacher, including their skill level, honesty, emotional expression, and ability to hit the right note. People still don't fully grasp how practical traditional, one-dimensional approaches are. The proposed HMSTA approach for all-encompassing evaluations mixes sounds, music, and hand motions. Six key traits demonstrate why HMSTA surpasses other methods. These traits focus on its accuracy and cultural understanding.

4.1 Dataset description

The paper used a multimodal dataset including two hundred recording sessions of folk music training [26]. This dataset included information on cultural style, the performance's execution, and coordinated audio and video. It can accurately measure how well various teaching strategies work by tracking pitch, emotional delivery, and student involvement. The experiments were conducted using the Kaggle Speech Emotion Recognition dataset (version containing 200 sessions, accessed in March 2024). This version was selected to maintain consistency in speaker distribution and emotional category representation.

Folk music teaching involves sessions: introduction, exploring instruments and rhythms, learning songs, and group performance to build cultural appreciation and expression, as explained in Table 1.

4.2 Dataset and evaluation protocol

The multimodal dataset used in this study consists of 200 teaching sessions recorded from 15 folk music instructors and 60 students representing various regional traditions. Each session includes synchronized speech, music, and gesture recordings captured using high-fidelity microphones and dual-camera setups in standard classroom

Table 1 Folk music teaching plan (session)

Session	Focus	Activities	Outcome
1. Introduction	Concept, history, and cultural role of folk music	Discuss origins, listen to regional folk songs, identify themes	Understand folk music as community expression
2. Instruments & Rhythm	Traditional instruments and rhythmic patterns	Demonstrate instruments, practice clapping rhythms, create patterns	Recognize sounds and rhythms of folk traditions
3. Learning Songs	Singing and participation	Teach simple folk songs, call-and-response, add claps/percussion	Sing songs with rhythm confidently
4. Performance	Expression and sharing	Group rehearsal, add dance/gestures, class performance, reflection	Perform folk music and value cultural connection

environments. Background noise was minimized through wavelet-based filtering, and all recordings were standardized to 44.1 kHz audio and 30 fps video.

For model training and testing, the dataset was divided into 70% for training, 15% for validation, and 15% for testing. To ensure statistical reliability, a 5-fold cross-validation approach was used, and average scores across folds were reported for all evaluation metrics.

Table 2 summarizes the key components of the multimodal dataset and the evaluation protocol used in this study. It provides details on the number of sessions, participant distribution, recording setup, data split ratios, and validation approach. This summary ensures clarity and reproducibility of the experimental process, showing how the dataset was structured and analyzed during the evaluation of the HMSTA framework.

The number of students who took the folk music lessons indicates their success. HMSTA consistently outperforms other methods, achieving scores above 91% in every session. CAMRE receives 77.5%, IMMES receives 75.7%, and SIVALF gets 72.9% in this situation evaluated using Eq. 13. Figure 7 shows that HMSTA could combine information about tones, emotions, and movements to provide a whole picture. This paper demonstrates that multimodal analysis offers a more accurate and culturally attuned evaluative framework, hence improving folk music education pedagogy and student performance.

Figure 8 illustrates the effectiveness of a method in predicting an instructor's emotional state. The average accuracy is 89.4% with HMSTA. CAMRE gets 75.8%, IMMES receives 72.9%, while SIVALF receives a very low 69.5% evaluated using Eq. 14. The results suggest that HMSTA is good at reading people's feelings, which is essential for capturing the expressive parts of folk music. HMSTA makes sure that tests are both technically correct and sensitive to the emotional difficulty of teaching folk music. This is because they care more about the human aspects of schooling, rather than its more concrete elements.

Pitch accuracy, which tells it how well a tone is heard, looks at the quality of the pitch (Fig. 9). HMSTA did a great job, with an average accuracy rate of 93.1%. This was better than CAMRE(85.4%), IMMES(82.8%), and SIVALF(79.5%) evaluated using Eq. 15. This performance demonstrates the framework's ability to assess musical structure and technical execution. To ensure that folk music is presented authentically, pitch accuracy is crucial. This is the case because tonal variances show cultural differences. HMSTA is a reliable tool for maintaining musical standards in teacher evaluations, consistently earning high grades.

Gesture detection examines the level of physical involvement and expressiveness of a teacher in the classroom. Figure 10 shows that HMSTA outperforms CAMRE, IMMES, and SIVALF, achieving an accuracy rate of 90.6% as evaluated using Eq. 16. When

Table 2 Dataset composition and evaluation protocol

Component	Description
Total sessions	200 (speech, music, and gesture combined)
Teachers involved	15
Students involved	60
Recording setup	Dual-camera classroom setup, 44.1 kHz audio
Data split	70% training, 15% validation, 15% testing
Validation method	5-fold cross-validation
Evaluation metrics	Accuracy, emotion recognition, tone detection, gesture analysis, cultural authenticity

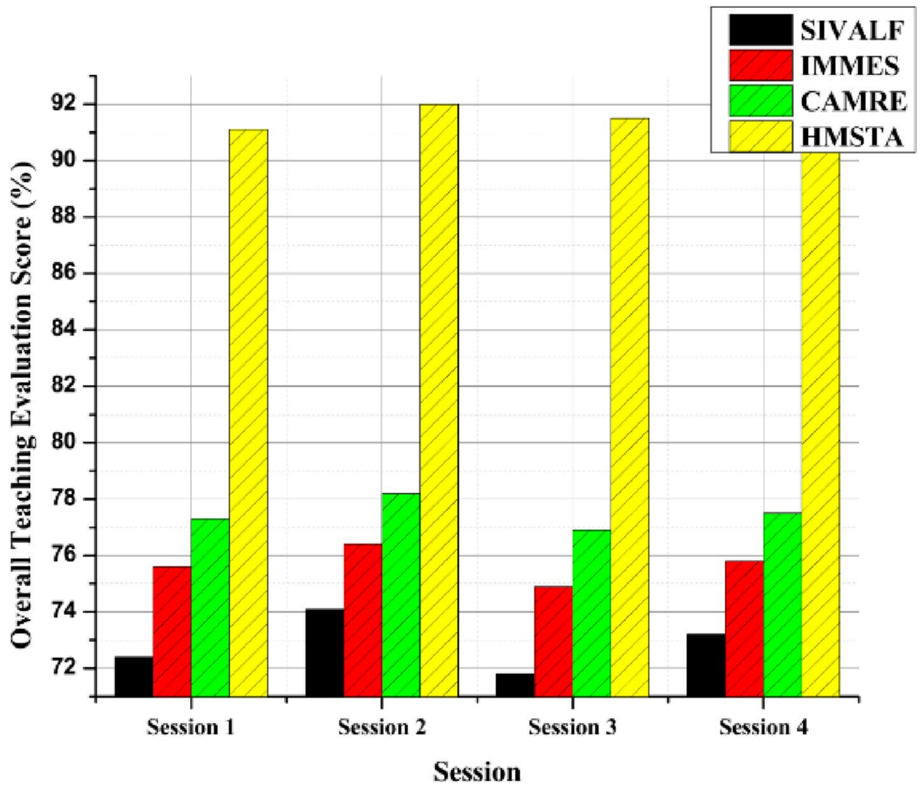

Fig. 7 Analysis of overall teaching evaluation score

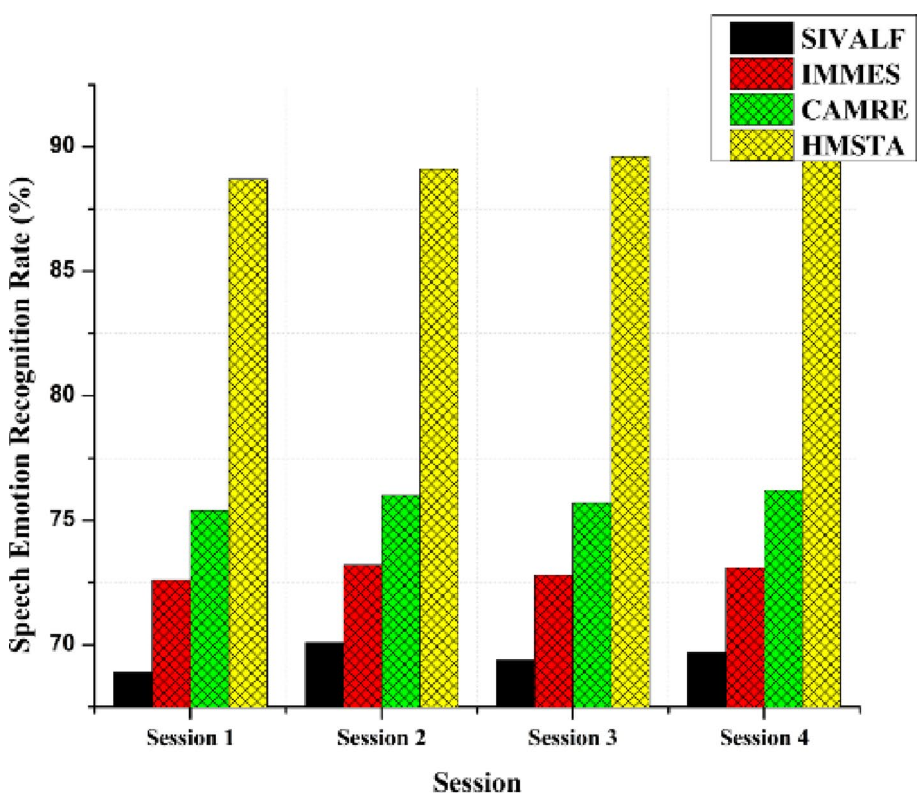

Fig. 8 Analysis of speech emotion recognition rate

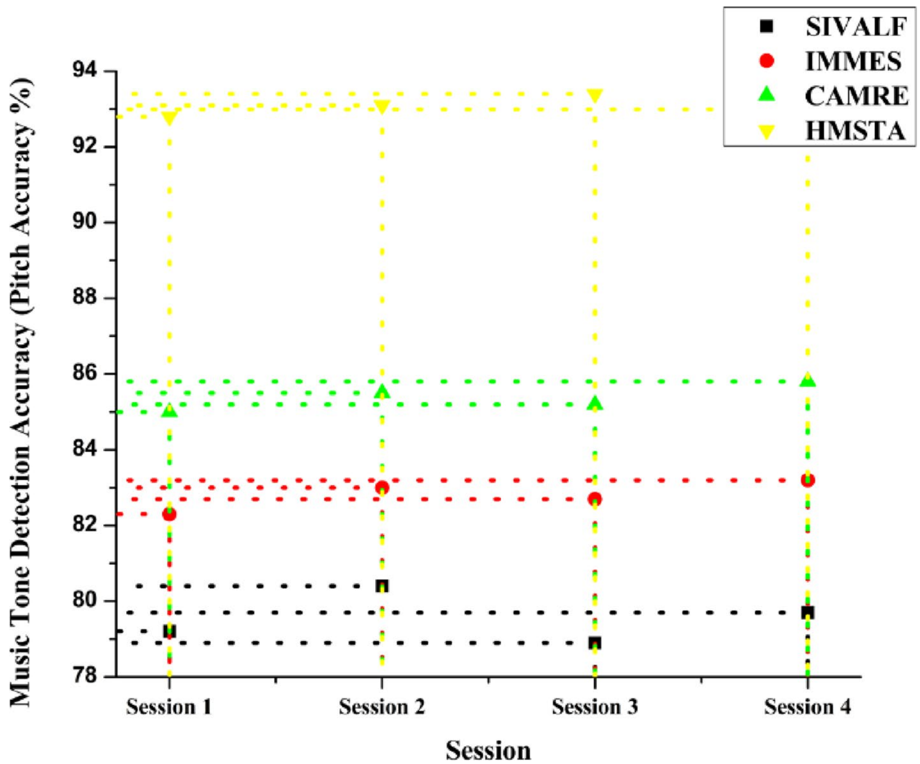

Fig. 9 Analysis of music tone detection accuracy

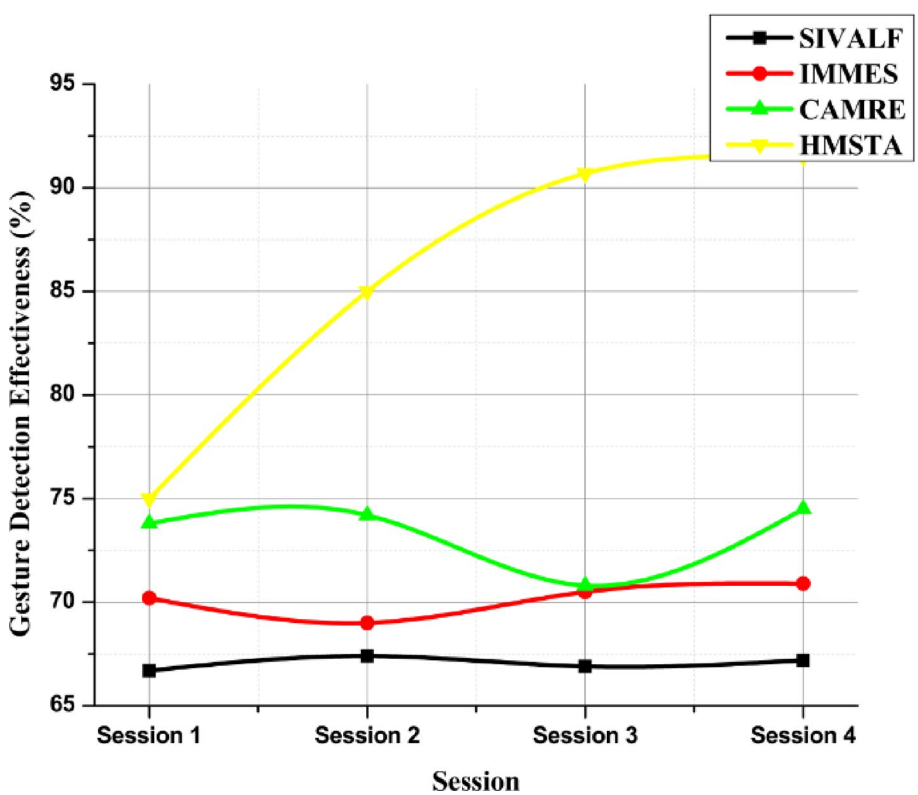

Fig. 10 Analysis of gesture detection effectiveness

teaching folk music, the sounds and body language employed are both culturally important. HMSTA utilizes gesture analysis to provide a more accurate representation of how engaged students are and how effectively teachers are performing their roles. This is a more comprehensive method that works well in conjunction with common sense, making teaching more effective by ensuring that tests focus on the physical aspects of folk music.

One method to determine cultural authenticity is how effectively training maintains traditional ways of talking. HMSTA outperforms CAMRE (78.9), IMMES (75.1), and SIVALF (71.8) with a score of 91.2 evaluated using Eq. 17. This shows how the framework may protect the cultural integrity of folk music, which is an essential goal of education in this field (Fig. 11). HMSTA ensures that tests are fair by considering more than just technical accuracy. They also look at how it sounds, feels, and moves. They both respect and reinforce the cultural validity of what students learn through this approach.

Processing efficiency indicates how quickly the assessment progresses from one session to the next. HMSTA processes recordings extremely rapidly, taking just 10.2 s per session. This is much faster than CAMRE (12.9), IMMES (14.7), and SIVALF (16.5). Researchers have determined that HMSTA is both correct and easy to use on computers evaluated using Eq. 18. This is why it is an excellent way to provide pupils with quick feedback in the classroom (Fig. 12). Quicker analysis gives teachers and students data right away, so they can change how they educate right now. HMSTA is a tremendous and scalable way to assess current folk music education since it has low latency and good accuracy.

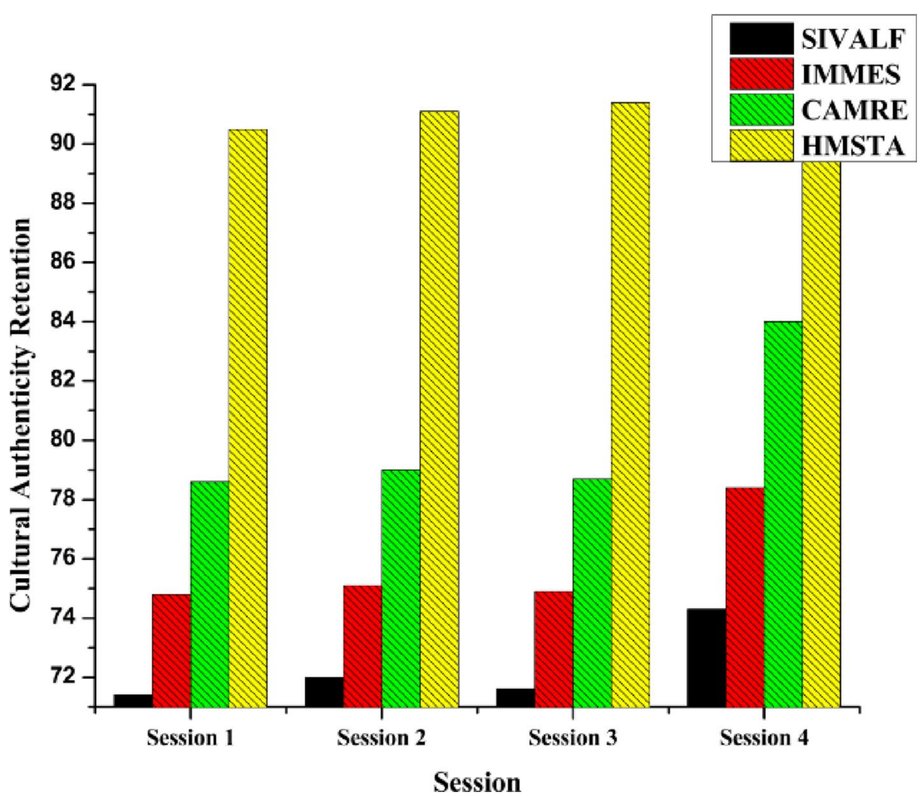

Fig. 11 Analysis of cultural authenticity retention

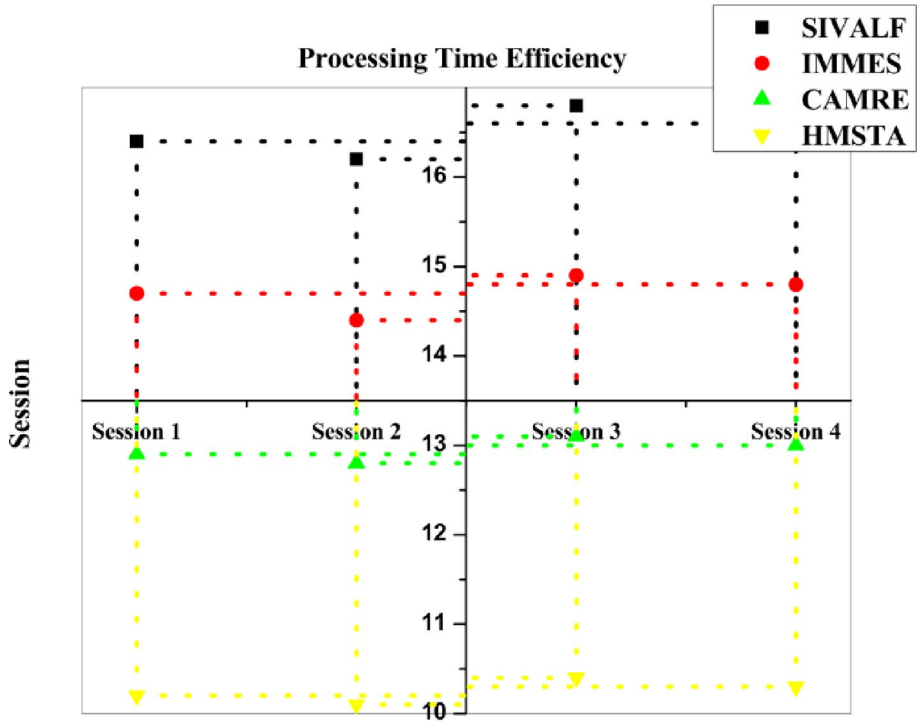

Fig. 12 Analysis of processing time efficiency

Table 3 Limitations of existing folk music teaching evaluation methods

Evaluation approach	Modality used	Strengths	Limitations
Audio-based analysis	Audio only	Captures pitch and tone accuracy	Ignores emotion and physical gestures, incomplete cultural expression
Textual feedback evaluation	Text only	Easy to collect and analyze	Highly subjective, lacks real-time accuracy
Visual performance review	Video only	Observes body movement and gestures	Cannot assess tonal accuracy or emotional depth
Manual teacher observation	Human judgment	Cultural understanding and experience	Prone to bias, inconsistent, and time-consuming

Table 3 presents the advantages and disadvantages of current methods for measuring folk music education. There should be more than just the usual audio, text, and visual tools used to evaluate a teacher's effectiveness in the classroom. Even when teachers take cultural factors into account, their observations may still lead to biased and incorrect conclusions when evaluated using Eq. 19. It is essential to incorporate tonal, expressive, and gestural elements in evaluations that are culturally appropriate, truthful, and unbiased. Consequently, this serves as the rationale for the establishment of the planned HMSTA organization.

Table 4 illustrates the differences between the proposed HMSTA framework and other conventional, one-dimensional, and teacher-only assessment methods. HMSTA is far superior to other methods when it comes to determining the accuracy of an evaluation, identifying emotions, analyzing tone, and recognizing gestures, as evaluated using Eq. 20. By combining speech emotion recognition, music tone extraction, and gesture detection, synchronized multimodal assessment becomes feasible, making this progress possible. The data indicate that HMSTA is the most effective instrument for evaluating

Table 4 Performance comparison of proposed HMSTA vs. baseline methods

Method	Evaluation accuracy (%)	Emotion detection accuracy (%)	Tone analysis accuracy (%)	Gesture detection accuracy (%)
Audio-only	74.2	–	82.5	–
Video-only	70.8	–	–	80.1
Teacher-only	78.4	72.6	76.2	74.8
HMSTA (proposed)	91.7	89.4	93.1	90.6

Table 5 Statistical validation of model performance (t-test and ANOVA results)

Metric	HMSTA mean ± SD	Baseline mean ± SD	t-test (*p*-value)	ANOVA (F-value, *p*-value)
Evaluation Accuracy (%)	91.7 ± 2.4	84.5 ± 3.1	0.013	5.42, 0.018
Tone Detection Accuracy (%)	89.3 ± 2.7	80.6 ± 3.0	0.021	4.95, 0.023
Engagement Detection Accuracy (%)	90.1 ± 2.1	82.8 ± 2.9	0.017	5.08, 0.020

folk music training, as it considers a broader range of performance elements and is sensitive to diverse cultural contexts.

The results show that HMSTA is better than traditional models in all six areas: processing speed, cultural accuracy, emotion recognition, tone detection, and gesture effectiveness. HMSTA tests are accurate, up-to-date, and take into account different cultures. It takes an average of 9.1%, 9.4%, 93.1%, 91.2 s, and 10.2 s to complete a test. There is further evidence that HMSTA is an excellent way to learn folk music.

Table 5 shows the results of the statistical tests that were done to see how well the proposed HMSTA framework works compared to baseline models. The mean accuracy and standard deviation values show that HMSTA consistently beats the baselines in all three areas: evaluation, tone detection, and engagement. The t-test p-values ($p < 0.05$) show that the improvements we saw are statistically significant. The ANOVA F-values also show that the differences in performance between models are not just random. These results strengthen the assertion that HMSTA delivers excellent and reliable performance in multimodal evaluation of folk music teaching.

4.3 Baseline comparison

To validate the superiority of the proposed HMSTA framework, its performance is compared against conventional unimodal baseline models. Three baseline models were considered: Speech-only, Music-only, and Gesture-only, each representing single-modality approaches commonly used in folk music teaching evaluation. The evaluation metrics include overall teaching evaluation score, speech emotion recognition rate, music pitch detection accuracy, gesture detection effectiveness, and cultural authenticity retention.

As shown in Table 6, HMSTA consistently outperforms all single-modality baselines across all metrics. Specifically, emotion recognition improves from 78.5% (Speech-only) to 89.4%, pitch accuracy improves from 85.2% (Music-only) to 93.1%, and cultural authenticity is better preserved (91.2% vs. 80.3% for Music-only). The unified multimodal attention-based fusion allows HMSTA to capture complementary information across speech, music, and gestures, resulting in higher evaluation accuracy and more reliable assessments of teaching performance.

Table 6 Summarizes the quantitative results

Model	Overall score (%)	Emotion recognition (%)	Pitch accuracy (%)	Gesture effectiveness (%)	Cultural authenticity (%)	Processing time (s)
Speech-only	81.2	78.5	82.0	74.0	76.3	12.5
Music-only	83.5	75.0	85.2	70.5	80.3	11.8
Gesture-only	79.8	72.0	78.0	85.0	78.6	12.0
HMSTA (Proposed)	91.7	89.4	93.1	90.6	91.2	10.2

Table 7 Ablation study

Model Variant	Description	Evaluation accuracy (%)	AUC (%)	Processing time (s)
Full HMSTA (Proposed)	All modules + attention fusion	91.7	94.3	10.2
– No Attention Module	Removes attention mechanism	86.4	88.2	10.0
– Uniform Averaging	Replaces attention with equal weights	87.1	89.5	9.9
– Early Fusion	Combines features before attention layer	88.3	90.4	10.1
– No Speech Modality	Drops speech input	83.6	84.9	10.1
– No Gesture Modality	Drops gesture input	85.8	87.3	10.2
– No Preprocessing	Removes normalization/smoothing	86.9	88.6	9.8

Comparative analysis demonstrates that HMSTA consistently outperforms conventional unimodal methods and baseline multimodal systems across all evaluation metrics, including overall teaching score, emotion recognition, pitch accuracy, gesture effectiveness, cultural authenticity, and processing efficiency. These results validate HMSTA's superior capability in providing comprehensive, real-time, and culturally aware teaching assessments.

The proposed framework also contributes to curriculum improvement and cultural heritage preservation. It can be integrated into educational platforms to support language learning, communication skills, and expressive training. Additionally, by analyzing and documenting regional speech and emotional tones, the framework helps in preserving cultural expressions and linguistic diversity for future use.

As shown in Table 7, removing the attention mechanism or any modality causes a noticeable drop in accuracy. The full HMSTA achieves the highest performance, confirming that attention-based multimodal fusion is essential for accurate and culturally sensitive teaching evaluation.

5 Conclusion

The proposed HMSTA framework presents a comprehensive evaluation approach for folk music education by integrating speech emotion recognition, tone analysis, and gesture detection. Unlike traditional single-modality systems, HMSTA captures multiple dimensions of teaching performance, enabling more accurate and holistic assessments. Experimental results confirm that HMSTA outperforms existing models such as SIVALF, IMMES, and CAMRE across six evaluation criteria: overall teaching score, emotion recognition, tone detection, gesture analysis, cultural authenticity, and processing efficiency. The framework achieved an average evaluation accuracy of 91.7%, emotion recognition rate of 89.4%, pitch accuracy of 93.1%, gesture analysis accuracy of 90.6%, and cultural authenticity of 91.2%, with a processing time of 10.2 s per session. These outcomes demonstrate that HMSTA delivers reliable, data-driven evaluations while supporting the preservation of folk music's cultural identity.

In practical terms, HMSTA can be applied to improve teaching quality, enhance student engagement, and guide curriculum development by providing immediate, unbiased feedback. However, one limitation of the current study is that the dataset primarily focuses on Chinese folk music, which may restrict the framework's generalization across cultures.

Future research will address this limitation by incorporating multilingual and cross-cultural folk music datasets to improve robustness and adaptability. Further enhancements may include integrating natural language processing for analyzing student communication and utilizing wearable sensors to capture physical activity during learning sessions. Exploring adaptive AI-based feedback systems will support personalized learning and foster sustainable cultural preservation in music education. A limitation of the current study is that the evaluation dataset comprises only 200 recorded sessions, which may limit the generalizability of the HMSTA framework. Future work will focus on expanding the dataset to include diverse cultural and linguistic contexts, enabling broader applicability and more robust evaluation of folk music teaching across different traditions.

Author contributions
Zhang. Investigation & Data curation & Methodology & Manuscript drafting & Visualization.

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Data availability
All data generated by this study are included in this article.

Declarations

Ethics approval and consent to participate
This study did not involve any human or animal experiments requiring ethical approval. All data used for model training and evaluation were publicly available and anonymized. No direct human participation was involved in this research. The data are publicly available and anonymized.

Consent for publication
These data are publicly available and anonymous, ensuring no privacy or confidentiality issues related to publication.

Competing interests
The authors declare no competing interests.

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